

EMBEDDED INTERNET SYSTEMS

Bricks to Build the Internet of Things and Other Dreams

Invisible little computers are here, there and everywhere today—and many of them are capable of connecting to the Internet as well. Have you ever wondered what are the specific requirements and challenges of making Internet-capable embedded systems? Let us go on a hop-on-hop-off ride into the bellies of these smart, little devices



In-dash computing device with internet connectivity via embedded 3G modem, hands-free phone functionality, Garmin navigation and mobile office productivity applications in truck cabin (Courtesy: www.fordworksolutions.com)

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At one time we had to think of Internet connectivity only from the perspective of desktop and laptop computers. Then it went on to personal digital assistants (PDAs) and other handhelds, followed by mobile phones, industrial instruments and machinery. Now it's even existent in printers, televisions and cars, and the future sees it in cameras, microwaves and washing

machines too!

What is the best possible way to make Internet-capable embedded systems for these innumerable devices? Do we need special processors? What kind of operating systems? How do you make the systems smaller, cooler and faster? How can we make them interoperable and secure? Which are the best interfaces and protocols?

As more and more people try to find the best answers to these questions, pledging their knowledge and

skills to the realisation of the 'Internet of Things'—a vision of connecting ambient objects and devices to monitor and control our environment, we see lots of brainwaves and lots of quizzical looks too. Of course, with so many developments around the world it could sometimes get difficult to keep track of where we are heading. In this article, we plan to ease some of those creased brows with an express overview of recent trends concerning embedded Internet systems.

Inside all things, small and big

"The area of networked embedded applications, termed as 'deeply embedded systems,' deals with embedded devices connected over a network. It has a multitude of sub areas such as home automation, industrial networks, vehicular automation and wireless sensor networks," begins Prof. K.R. Anupama, associate professor & head of the department-electrical & electronics engineering, BITS, Pilani, K.K. Birla Goa Campus. Intel terms this as the fourth phase in the evolution of the Internet; the first being the connection of mainframes, the second being e-commerce and e-mail revolution, and the third being social networking. With embedded systems getting more advanced and so many devices waiting to be connected to the Internet, there is no end to the